Stable diffusion 2.1 demo deployment

Export or download the pre-compiled model context file

There are 2 ways to get the model files.

- Method 1 - Export by QC's python Script.

  Clone https://github.com/quic/ai-hub-models on PC and export the pre-compiled model.

```cmd
D:\py\venv\Scripts\Activate.ps1
 python -m qai_hub_models.models.stable_diffusion_v2_1.export --target-runtime precompiled_qnn_onnx  --device "QCS9075" --fetch-static-assets v0.39.1
```

- Method 2 - Download from Huggingface

  Download from Huggingface, and find the corresponding version.

Pay attention to the model version, such as v0.39.1. We need to find the QNN SDK version information corresponding to this model from the tool-versions.yaml on the download page or from the model context file, and the QNN SDK version on QCS9100 must be at least consistent with it.

If the version in tool-versions.yaml is `qairt: 2.38.0.250901140452_125126-auto`, the target machine must have at least V2.38.0 installed.

> Version information can also be found in the model file under Linux using the strings command:
>
> ```cmd
> strings Stable-Diffusion-v2.1_text_encoder_w8a16.bin | grep "v2."
> v2.39.0.250925215840_163802-auto
> Vrv2.39.0.250925215840_163802-auto.fcaeba5a50
> Vrv2.39.0.250925215840_163802-auto.fcaeba5a50
> ```
>
> Additionally, we can also build/convert the model context files by ourselves following https://qpm.qualcomm.com/#/main/tools/details/Tutorial_for_Stable_Diffusion_2_1_Compute

Three model context files will be downloaded: text-encoder, unet and vae :

```cmd
Stable-Diffusion-v2.1_text_encoder_w8a16.bin
Stable-Diffusion-v2.1_unet_w8a16.bin
Stable-Diffusion-v2.1_vae_w8a16.bin
```

Install QNN SDK

Select the right version of QNN SDK as above, and install it from https://qpm.qualcomm.com/#/main/tools/details/Qualcomm_AI_Runtime_SDK on a host PC. After the SDK is installed, copy the SDK binaries and binaries to the QCS9075/QCS9100 target device as folowing structure.

```
代码块
2.39.0.250926/
|-- bin
|   `-- aarch64-oe-linux-gcc11.2
|       |-- genie-app
|       |-- genie-t2e-run
|       |-- genie-t2t-run
|       |-- qnn-context-binary-generator
|       |-- qnn-context-binary-utility
|       |-- qnn-net-run
|       |-- qnn-platform-validator
|       |-- qnn-profile-viewer
|       |-- qnn-throughput-net-run
|       |-- qtld-net-run
|       |-- snpe-diagview
|       |-- snpe-net-run
|       |-- snpe-parallel-run
|       |-- snpe-platform-validator
|       `-- snpe-throughput-net-run
`-- lib
    |-- aarch64-oe-linux-gcc11.2
    |   |-- libGenie.so
    |   |-- libPlatformValidatorShared.so
    |   |-- libQnnChrometraceProfilingReader.so
    |   |-- libQnnCpu.so
    |   |-- libQnnCpuNetRunExtensions.so
    |   |-- libQnnDsp.so
    |   |-- libQnnDspNetRunExtensions.so
    |   |-- libQnnDspV66CalculatorStub.so
    |   |-- libQnnDspV66Stub.so
    |   |-- libQnnGenAiTransformer.so
    |   |-- libQnnGenAiTransformerCpuOpPkg.so
```

```
32      |   |-- libQnnGenAiTransformerModel.so
33      |   |-- libQnnGpu.so
34      |   |-- libQnnGpuNetRunExtensions.so
35      |   |-- libQnnGpuProfilingReader.so
36      |   |-- libQnnHta.so
37      |   |-- libQnnHtaNetRunExtensions.so
38      |   |-- libQnnHtp.so
39      |   |-- libQnnHtpNetRunExtensions.so
40      |   |-- libQnnHtpOptraceProfilingReader.so
41      |   |-- libQnnHtpPrepare.so
42      |   |-- libQnnHtpProfilingReader.so
43      |   |-- libQnnHtpV68CalculatorStub.so
44      |   |-- libQnnHtpV68Stub.so
45      |   |-- libQnnHtpV69CalculatorStub.so
46      |   |-- libQnnHtpV69Stub.so
47      |   |-- libQnnHtpV73CalculatorStub.so
48      |   |-- libQnnHtpV73Stub.so
49      |   |-- libQnnHtpV75CalculatorStub.so
50      |   |-- libQnnHtpV75Stub.so
51      |   |-- libQnnHtpV79CalculatorStub.so
52      |   |-- libQnnHtpV79Stub.so
53      |   |-- libQnnIr.so
54      |   |-- libQnnJsonProfilingReader.so
55      |   |-- libQnnModelDlc.so
56      |   |-- libQnnSaver.so
57      |   |-- libQnnSystem.so
58      |   |-- libQnnTFLiteDelegate.so
59      |   |-- libSNPE.so
60      |   |-- libSnpeDspV66Stub.so
61      |   |-- libSnpeHta.so
62      |   |-- libSnpeHtpPrepare.so
63      |   |-- libSnpeHtpV68CalculatorStub.so
64      |   |-- libSnpeHtpV68Stub.so
65      |   |-- libSnpeHtpV73CalculatorStub.so
66      |   |-- libSnpeHtpV73Stub.so
67      |   |-- libSnpeHtpV75CalculatorStub.so
68      |   |-- libSnpeHtpV75Stub.so
69      |   |-- libSnpeHtpV79CalculatorStub.so
70      |   |-- libSnpeHtpV79Stub.so
71      |   |-- libcalculator.so
72      |   |-- libhta_hexagon_runtime_qnn.so
73      |   `-- libhta_hexagon_runtime_snpe.so
74      `-- hexagon-v73
75          `-- unsigned
76              |-- libCalculator_skel.so
77              |-- libQnnHtpV73.so
78              |-- libQnnHtpV73QemuDriver.so
```

```
            |-- libQnnHtpV73Skel.so
            |-- libQnnSaver.so
            |-- libQnnSystem.so
            |-- libSnpeHtpV73Skel.so
            |-- libqnnhtpv73.cat
            `-- libsnpehtpv73.cat
```

Python setup

> If your device has pip3, please ignore this section.

Since only a minimal python3 is integrated in our LE (Open Embedded) system, we have to install a full-fledged Python build, and pip3 is required as well.

Download python3.12.9 and python3.12.9-lib rpm package and extract files from them on a Linux host:

cmd on host

```
wget https://yum.oracle.com/repo/OracleLinux/OL9/appstream/aarch64/getPackage/python3.12-3.12.9-1.el9.aarch64.rpm
wget https://yum.oracle.com/repo/OracleLinux/OL9/appstream/aarch64/getPackage/python3.12-libs-3.12.9-1.el9.aarch64.rpm
#extract files
rpm2cpio python3.12-3.12.9-1.el9.aarch64.rpm | cpio -idmv
rpm2cpio python3.12-libs-3.12.9-1.el9.aarch64.rpm | cpio -idmv

#a usr directory will be extracted,copy it on the device
scp -r usr/bin/* usr@device-ip:/usr/bin/
scp -r usr/include/* usr@devcie-ip:/usr/include/
scp -r usr/share/* usr@device-ip:/usr/share
scp -r usr/lib/*  usr@device-ip:/usr/lib
scp -r usr/lib64/* usr@device-ip:/usr/lib
```

Link a lib64 directory on the device

cmd

```
ln -s /usr/lib /usr/lib64
ldconfig
```

Install pip3 on the device

```
wget https://bootstrap.pypa.io/get-pip.py
python3 get-pip.py
#check pip3
pip3 --version
```

Create a virtual environment(venv) and activate it:

```cmd
python3 -m venv sd21-project-env
source sd21-project-env/bin/activate
```

Install python modules by the runner script in the venv

```cmd
pip install diffusers==0.35.2 \
            numpy==2.3.5 \
            pillow==12.0.0 \
            tokenizers==0.22.1 \
            torch==2.9.1 \
            transformers==4.57.1 \
            accelerate \
            transfer \
            -i https://pypi.tuna.tsinghua.edu.cn/simple #optional for China mainland

```

Run Generative script

Clone or copy this project on the device

```cmd
#In China mainland, use a huggingface mirror site instead
export HF_ENDPOINT=https://hf-mirror.com
export HF_HUB_ENABLE_HF_TRANSFER=1  #Optional, may speed up the download in first execution
                                    #but has a high chance to be reject by hf-mirror.com

python3 sd21_qnn_linux.py --prompt "A kitten is practicing martial arts" --steps 20 --seed 1 --guidance 7.5 --output sd21_qnn.png
```

> The first execution will take a long time to download some resources from huggingface or the mirror site.

> The above steps were executed and verified on LE. We also tried on LU, but failed due to the integrity of the OS and the support level of the SDK

LU

Issue List

Issue	Solution & Workaround
[Both LE and LU]Failed to fetch something from Hugingface like: 代码块 1 '(MaxRetryError("HTTPSConnectionPool(host='huggingface.co', port=443): Max retries exceeded with url: /stabilityai/stable-diffusion-2-1-base/resolve/main/unet/config.json (Caused by NewConnectionError('<u	There are multiple possibilities causing this issue. 1. The resource was removed from Huggingface. In this case we need to use community d2-community/stable-diffusion-2-1-base intead of official stabilityai/stable-diffusion-2-1-base for time embedding (updating python module[diffusers transformers]may help, but we have not tried) 2. China mainland has no access to Huggingface. It need a VPN or to execute the generating out of China mainland. Another recommended way is using a mirror site of huggingface by `export HF_ENDPOINT=https://hf-mirror.com` Tokenizer json file clip-vit-large-patch14-tokenizer.json will be downloaded from openai/clip-vit-large-patch14. Different python tokenizer version may download different json file, and they may be incompatible with each other.

```
rllib3.connection.Verifiedhttpsconnection object at 0x7f63e93610>: Failed to establish a new connection: [Errno 101] Network is unreachable'))"), '(Request ID: 4e87f358-0d0a-41dc-87a5-ba4cf79d15ce)')' thrown while requesting HEAD https://huggingface.co/stabilityai/stable-diffusion-2-1-base/resolve/main/unet/config.json
Retrying in 1s [Retry 1/5].
...
...
```

[LU]Running with AP side library from aarch64-ubuntu-gcc9.4:

代码块

```
      1.0ms [VERBOSE] <V> Config not passed. Loading default config
      1.0ms [VERBOSE] <V> Setting default value for unsigned PD usage
      1.0ms [ ERROR ] <E> HtpRuntimeStub: Unsupported SoC 0
      1.0ms [ ERROR ] <E> failed to initialize skel/stub file path
      1.2ms [WARNING] <W> Failed in loading stub: libQnnHtp:
```

V73 dsp has no corresponding stub for ubuntu for this SDK. As a workaround, we use aarch64-oe-linux-gcc9.4 lib/bin even thought the platform is ubuntu-24.04-QCS9075

```
cannot open shared object file: No such file or directory
      1.2ms [WARNING] <W> Failed to create transport instance: 4000
      1.2ms [ ERROR ] <E> Failed to create transport for device, error: 4000
      1.2ms [ ERROR ] <E> Failed to load skel, error: 4000
      1.2ms [VERBOSE] <V> exits device initialization with 14001
```

[LU]Memory issue(**1002**)

代码块

```
      1.2ms [VERBOSE] <V> Effective cdsp_id is: 0, Session_id is: 0 for DeviceId: 0, CoreId: 0, pdId: 0
      1.2ms [ ERROR ] <E> Fail to get effective domain id from rpc with DeviceId 0 coreId 0 pdId 0
      1.2ms [ ERROR ] <E> error in creation of transport instance
      1.2ms [WARNING] <W> Failed to create transport instance: 1002
      1.2ms [ ERROR ] <E> Failed to create transport for device, error: 1002
```

This is a kernel issue:

By `journalctl -k | grep dsp -i` , we find kernel error log:

no reserved DMA memory for FASTRPC

代码块

```
Jul 02 14:04:33 Thundercomm kernel: OF: reserved mem: initialized node secure-cdsp-region, compatible id shared-dma-pool
Jul 02 14:04:33 Thundercomm kernel: OF: reserved mem: 0x00000000fdc00000..0x00000000ffbfffff (32768 KiB) map reusable secure-cdsp-region
Jul 02 14:04:33 Thundercomm kernel: OF: reserved mem: 0x0000000094a00000..0x00000000951fffff (8192 KiB) nomap non-reusable adsp-rpc-remote-heap@94a00000
Jul 02 14:04:33 Thundercomm kernel: OF: reserved mem: 0x0000000095900000..0x00000000976fffff (30720 KiB) nomap non-reusable pil-adsp@95900000
```

```
      1.2ms [ ERROR ] <E> Failed to load skel, error: 1002
      1.2ms [VERBOSE] <V> exits device initialization with 14001
      1.2ms [ ERROR ] <E> Transport layer setup failed: 14001
```

```
Jul 02 14:04:33 Thundercomm kernel: OF: reserved mem: 0x0000000097700000..0x000000009777ffff (512 KiB) nomap non-reusable q6-adsp-dtb@97700000
Jul 02 14:04:33 Thundercomm kernel: OF: reserved mem: 0x0000000097780000..0x00000000977fffff (512 KiB) nomap non-reusable q6-gdsp0-dtb@97780000
Jul 02 14:04:33 Thundercomm kernel: OF: reserved mem: 0x0000000097800000..0x00000000995fffff (30720 KiB) nomap non-reusable pil-gdsp0@97800000
Jul 02 14:04:33 Thundercomm kernel: OF: reserved mem: 0x0000000099600000..0x000000009b3fffff (30720 KiB) nomap non-reusable pil-gdsp1@99600000
Jul 02 14:04:33 Thundercomm kernel: OF: reserved mem: 0x000000009b400000..0x000000009b47ffff (512 KiB) nomap non-reusable q6-gdsp1-dtb@9b400000
Jul 02 14:04:33 Thundercomm kernel: OF: reserved mem: 0x000000009b480000..0x000000009b4fffff (512 KiB) nomap non-reusable q6-cdsp0-dtb@9b480000
Jul 02 14:04:33 Thundercomm kernel: OF: reserved mem: 0x000000009b500000..0x000000009d2fffff (30720 KiB) nomap non-reusable pil-cdsp0@9b500000
Jul 02 14:04:33 Thundercomm kernel: OF: reserved mem: 0x000000009d380000..0x000000009d3fffff (512 KiB) nomap non-reusable q6-cdsp1-dtb@9d380000
Jul 02 14:04:33 Thundercomm kernel: OF: reserved mem: 0x000000009d400000..0x000000009f1fffff (30720 KiB) nomap non-reusable pil-cdsp1@9d400000
Jul 02 14:04:33 Thundercomm kernel: platform soc@0:qcom,dma-
```

```
heaps:qcom,secure_cdsp: assigned reserved
 memory node secure-cdsp-region
Jul 02 14:04:33 Thundercomm kernel:
qcom_dma_heap_probe: DMA-BUF Heap:
Created qcom,cma-secure-cdsp
Jul 02 14:04:35 Thundercomm kernel:
remoteproc remoteproc0: Booting fw image
qcom/sa8775p/gpdsp0.mbn, size 5836800
Jul 02 14:04:35 Thundercomm kernel:
remoteproc remoteproc1: Booting fw image
qcom/sa8775p/gpdsp1.mbn, size 5808128
Jul 02 14:04:35 Thundercomm kernel:
remoteproc remoteproc2: Booting fw image
qcom/sa8775p/cdsp0.mbn, size 3080192
Jul 02 14:04:35 Thundercomm kernel:
remoteproc remoteproc3: Booting fw image
qcom/sa8775p/cdsp1.mbn, size 3084288
Jul 02 14:04:35 Thundercomm kernel:
remoteproc remoteproc4: Booting fw image
qcom/sa8775p/adsp.mbn, size 6647808
Jul 02 14:04:35 Thundercomm kernel:
qcom,fastrpc 20c00000.remoteproc:glink-
edge.fastrpcglink-apps-dsp.-1.-1: no
reserved DMA memory for FASTRPC
Jul 02 14:04:35 Thundercomm kernel:
qcom,fastrpc 21c00000.remoteproc:glink-
edge.fastrpcglink-apps-dsp.-1.-1: no
reserved DMA memory for FASTRPC
Jul 02 14:04:35 Thundercomm kernel:
qcom,fastrpc 30000000.remoteproc:glink-
edge.fastrpcglink-apps-dsp.-1.-1: no
reserved DMA memory for FASTRPC
Jul 02 14:04:35 Thundercomm kernel:
fastrpc: fastrpc_setup_service_locator:
pdr_add_lookup enabled for avs/audio
(audio_pdr_adsp, msm/adsp/audio_pd)
Jul 02 14:04:35 Thundercomm kernel:
fastrpc: fastrpc_setup_service_locator:
pdr_add_lookup enabled for tms/servreg
(sensors_pdr_adsp, msm/adsp/sensor_pd)
Jul 02 14:04:40 Thundercomm kernel:
qcom,fastrpc 26300000.remoteproc:glink-
edge.fastrpcglink-apps-dsp.-1.-1: failed
to create endpoint
Jul 02 14:04:40 Thundercomm kernel:
qcom,fastrpc: probe of
26300000.remoteproc:glink-
```

```
edge.fastrpcglink-apps-dsp.-1.-1 failed
with error -12
Jul 02 14:04:42 Thundercomm kernel:
qcom,apr 30000000.remoteproc:glink-
edge.adsp_apps.-1.-1: Adding APR/GPR dev:
 gprsvc:service:2:1
Jul 02 14:04:42 Thundercomm kernel:
qcom,fastrpc 30000000.remoteproc:glink-
edge.fastrpcglink-apps-dsp.-1.-1:
fastrpc_pdr_cb: msm/ads/audio_pd
(audio_pdr_adsp) is up for PDR on adsp
Jul 02 14:04:42 Thundercomm kernel:
qcom,apr 30000000.remoteproc:glink-
edge.adsp_apps.-1.-1: Adding APR/GPR dev:
 gprsvc:service:2:2
Jul 02 14:04:55 Thundercomm kernel:
qcom,fastrpc 26300000.remoteproc:glink-
edge.fastrpcglink-apps-dsp.-1.-1: no
reserved DMA memory for FASTRPC
Jul 02 14:04:55 Thundercomm kernel:
qcom,fastrpc 2a300000.remoteproc:glink-
edge.fastrpcglink-apps-dsp.-1.-1: no
reserved DMA memory for FASTRPC
Jul 02 14:06:55 Thundercomm kernel:
UBSAN: array-index-out-of-bounds in
/home/thundersoft/SSD/Qcs9100_DevKit_Ubun
tu_X05/linux/noble/sound/soc/qcom/qdsp6/q
6apm-lpass-dais.c:242:65
Jul 02 14:06:55 Thundercomm kernel:
UBSAN: array-index-out-of-bounds in
/home/thundersoft/SSD/Qcs9100_DevKit_Ubun
tu_X05/linux/noble/sound/soc/qcom/qdsp6/q
6apm-lpass-dais.c:149:31
Jul 02 14:06:55 Thundercomm kernel:
UBSAN: array-index-out-of-bounds in
/home/thundersoft/SSD/Qcs9100_DevKit_Ubun
tu_X05/linux/noble/sound/soc/qcom/qdsp6/q
6apm-lpass-dais.c:156:21
```

Key conflict points:

In the device tree, "adsp - rpc - remote - heap" has been defined.

The driver reports an error: "no reserved DMA memory for FASTRPC".

	This strongly indicates that the FastRPC driver fails to find the reserved memory area it should use. In other words, the device tree node of the FastRPC driver does not correctly reference this reserved memory node.

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